MD. NAZMUL ISLAM

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RESEARCH INTERESTS

Veterinary epidemiology; Bayesian quantitative risk assessment of transboundary and zoonotic animal diseases; infectious disease transmission dynamics; diagnostic-test evaluation and surveillance under uncertainty; livestock health and One Health in South Asia and other low- and middle-income countries.

EDUCATION

Master of Science in Medicine April 2025 – Present
Department of Medicine, Faculty of Veterinary Science, Bangladesh Agricultural University, Mymensingh
CGPA: 3.80 / 4.00 (first semester)
Thesis: *Bayesian Quantitative Risk Assessment of African Swine Fever Spread among Scavenging Pig Populations in Bangladesh.*

Doctor of Veterinary Medicine (DVM) January 2019 – January 2025
Faculty of Veterinary Science, Bangladesh Agricultural University, Mymensingh
CGPA: 3.49 / 4.00 (WES ICAP U.S.-equivalent GPA: 3.58 / 4.00)

PUBLICATIONS

Author name in bold. Nine published (three as first/equal-first author); five under review (two as first/equal-first author).

Peer-Reviewed Journal Articles — First or Equally Contributing First Author

1. Kibria, A.M., **Islam, M.N.,** Alam, M.S., Adhikari, B.J., Islam, S., Rafeh, A.A., Hasan, R.S.M., Rahman, M.S., & Rahman, A.K.M.A. (2026). Estimated true prevalence and associated risk factors of bovine paratuberculosis antibodies in dairy herds in Bangladesh. *PLOS ONE.* https://doi.org/10.1371/journal.pone.0334683
   Equally contributing first author *— study design, Bayesian latent-class analysis, manuscript writing.*
2. Mahmud Sajeeb, M.S., Alam, M.S., **Islam, M.N.,** Islam, M.M., Adhikari, B.J., Islam, S., Rahman, M.S., & Rahman, A.K.M.A. (2025). Prevalence and risk factors of bovine viral diarrhea virus antibodies in dairy herds of Bangladesh. *Veterinary Sciences, 12(8), 739.* https://doi.org/10.3390/vetsci12080739
   Equally contributing first author *— study design, field sampling, ELISA, analysis, manuscript writing.*
3. Saha, U.K., **Islam, M.N.,** Chaudhary, P., Hassan Alif, M.K., Mim, S.N., & Rahman, M.S. (2026). Age-related antibody kinetics and haematological safety after immunisation with a locally prepared heat-inactivated *Brucella abortus* vaccine candidate in Bangladeshi crossbred dairy cattle. *Bulgarian Journal of Veterinary Medicine* (online first). https://doi.org/10.15547/bjvm.2026-0070
   Equally contributing first author *— experimental design, ELISA and haematology, manuscript writing.*

Peer-Reviewed Journal Articles — Co-Author

4. Alam, M.S., **Islam, M.N.,** Adhikari, B.J., Islam, S., Mahmud Hasan, R.S., Rahman, M.S., Ariful Islam, M., Aktaruzzaman, M., Meletis, L., Kostoulas, P., & Rahman, A.K.M.A. (2026). Bayesian hierarchical mixture modelling to derive probabilistic iELISA thresholds for bovine brucellosis in endemic dairy systems. *PLOS ONE, 21(7), e0347719.* https://doi.org/10.1371/journal.pone.0347719
   Data curation, formal analysis, methodology, software, and writing — original draft.
5. Alam, M.S., **Islam, M.N.,** Adhikari, B.J., Islam, S., Mahmud, R.S., Rahman, M.S., Ariful Islam, M., Aktaruzzaman, M., & Rahman, A.K.M.A. (2026). Risk factors for latent and active brucellosis in dairy cattle: a Bayesian mixed-effects multinomial regression analysis in large-scale Bangladeshi herds. *Animal Diseases, 6, 26.* https://doi.org/10.1186/s44149-026-00234-w
   Data curation, Bayesian mixed-effects multinomial modelling, original draft.
6. Alam, M.S., **Islam, M.N.,** Adhikari, B.J., Islam, S., Rahman, M.S., Islam, M.A., Aktaruzzaman, M., & Rahman, A.K.M.A. (2026). Economic burden and cost-effectiveness of brucellosis control strategies in large commercial dairy herds in Bangladesh: a Bayesian decision-analytic study. *Preventive Veterinary Medicine*. https://doi.org/10.1016/j.prevetmed.2026.107014
   Data curation, formal analysis, methodology, software, original draft.

7. Aktaruzzaman, M., Ariful Islam, M., Shanta, S.A., Tahomina Akter, M., Hossain, M.T., **Islam, M.N.,** Alam, M.S., & Rahman, A.K.M.A. (2026). Prevalence and cow-level risk factors for mild and severe subclinical mastitis in intensive and extensive dairy systems in Bangladesh. *Veterinary Research Communications, 50,* 447. https://doi.org/10.1007/s11259-026-11387-x
8. Abu Saif, M., **Islam, M.N.,** Rahman, A.K.M.A., Ferdous, M.J., Islam, R., & Bari, F.Y. (2026). Integrated immune and protein profiles predict growth in preweaning lambs independent of feeding regimen and maternal oxytocin administration. *Iraqi Journal of Veterinary Sciences, 40(2), 397–403.* https://doi.org/10.33899/ijvs.2026.168586.4671
9. Abu Saif, M., **Islam, M.N.,** Ferdous, M.J., Rahman, A.K.M.A., & Bari, F.Y. (2026). Early-life feeding and management interventions on lamb survivability. *Bulgarian Journal of Veterinary Medicine* (online first). https://doi.org/10.15547/bjvm.2026-0050

Manuscripts Under Review — First or Equally Contributing First Author

1. Ferdous, M.J., **Islam, M.N.,** Alam, M.S., Saif, M.A., Rahman, M.A., Aktaruzzaman, M., Rahman, M.S., & Rahman, A.K.M.A. (2026). Age-stratified transmission dynamics of bovine brucellosis in large commercial dairy herds in Bangladesh: a hierarchical Bayesian catalytic analysis. *Epidemics.* **[Equally contributing first author]**
2. **Islam, M.N.,** Haque, M.Z., Alam, M.S., Islam, S., Adhikari, B.J., Rahman, M.S., & Rahman, A.K.M.A. (2026). Bayesian estimation of the true prevalence of *Coxiella burnetii* milk antibodies and associated risk factors in dairy herds in Bangladesh. *Veterinary Medicine and Science.* **[First author]**

Manuscripts Under Review — Co-Author

3. Anwarul Islam, **Islam, M.N.,** Ferdous, M.J., Alam, M.S., Rahman, M.A., Mahmud, R.S., Alam, M.M., & Rahman, A.K.M.A. (2026). A probabilistic decision-analytic model of the economic burden and strategic control of ruminant fascioliasis in Bangladesh. *Veterinary Parasitology: Regional Studies and Reports.*
4. Aktaruzzaman, M., Ariful Islam, M., Shanta, S.A., Tahomina Akter, M., Hossain, M.T., Alam, M.S., **Islam, M.N.,** Meletis, L., Kostoulas, P., & Rahman, A.K.M.A. (2025). Bayesian Gaussian mixture modelling to determine somatic cell count thresholds for subclinical mastitis diagnosis in dairy cows. *The Journal of Veterinary Medical Science.*
5. Aktaruzzaman, M., Tahomina Akter, M., Mazhar Islam, M., Ariful Islam, M., Hossain, M.T., Alam, M.S., **Islam, M.N.,** & Rahman, A.K.M.A. (2025). Economic impact of subclinical mastitis in Bangladeshi crossbred dairy cattle. *Preventive Veterinary Medicine.*

Manuscripts in Preparation

1. **Islam, M.N.,** Ferdous, M.J., Ema, A.A., Haque, M.Z., Alam, M.S., Rahman, M.S., & Rahman, A.K.M.A. Herd-level force of infection of *Coxiella burnetii* in large commercial dairy herds in Bangladesh: a hierarchical Bayesian serocatalytic analysis with concurrent *Neospora caninum* serology. **[First author]**

CONFERENCE PRESENTATIONS & PROCEEDINGS

1. **Islam, M.N.,** Ferdous, M.J., Islam, S., Alam, M.S., Rahman, M.S., & Rahman, A.K.M.A. (2026). Prevalence of *Neospora caninum* antibodies and associated risk factors in dairy herds in Bangladesh. *Oral presentation,* 2nd International Online Conference on Veterinary Sciences (IOCVS 2026), MDPI, Basel, Switzerland. https://sciforum.net/paper/33349
2. **Islam, M.N.,** Alam, M.S., Islam, S., Adhikari, B.J., & Rahman, A.K.M.A. (2025). Feasibility of on-farm somatic cell count screening for subclinical mastitis: prevalence and risk factors in a commercial dairy herd in Bangladesh. *Oral presentation,* 1st International Online Conference on Veterinary Sciences (IOCVS 2025), MDPI, Basel, Switzerland. https://sciforum.net/paper/view/27422

RESEARCH EXPERIENCE

MS Thesis Research · Dept. of Medicine, BAU (Supervisor: Prof. Dr. A.K.M. Anisur Rahman) Apr 2025 – Present

- Building a stochastic risk model of African swine fever (genotype II) spread among scavenging and nomadic pigs, following the WOAH framework: release via pig trade, nomadic flocks, cross-border movement and pork products; exposure via commingling, swill feeding and fomites.
- Estimating within-herd R_0 from daily outbreak data with Bayesian chain-binomial models (MCMC; WAIC comparison and model averaging), and calibrating expert priors to surveillance records by Bayesian melding.
- Simulating undetected spread as a herd-level branching process and ranking risk drivers and control options by Spearman and partial rank correlation.

Research Assistant · Epidemiology & Preventive Medicine Lab, Dept. of Medicine, BAU Jan 2023 – Present

- Analysed herd-level serological data from large commercial dairy herds for brucellosis, paratuberculosis, Q fever, neosporosis, bovine viral diarrhoea and subclinical mastitis.
- Wrote Bayesian models in R and Stan: latent-class true prevalence without a gold standard, hierarchical catalytic force-of-infection models, mixture-model test cut-offs, mixed-effects multinomial regression and decision-analytic cost-effectiveness models.
- Took part in farm sampling; ran indirect ELISA, DNA extraction and PCR; trained junior lab members in study design and laboratory procedures.
- Worked with international Bayesian diagnostic-test methodologists (P. Kostoulas, L. Meletis) on the threshold-modelling studies.

Independent Research Lead · Global human brucellosis epidemiology programme Mar 2026 – Present

- Initiated and lead a ten-member team synthesising global evidence on human brucellosis with Bayesian meta-analysis, spatiotemporal mapping and a One Health framing.

TECHNICAL & RESEARCH SKILLS

Bayesian modelling: R (rstan, brms, tidyverse, ggplot2) and Stan; MCMC fitting, WAIC model comparison and model averaging; hierarchical catalytic (force-of-infection) and chain-binomial transmission models (R_0); latent-class and mixture models; mixed-effects and multivariable regression; diagnostic-test evaluation

Risk assessment & simulation: WOAH import risk analysis framework; stochastic Monte Carlo risk models; Bayesian melding of expert priors; branching-process outbreak simulation; sensitivity analysis (Spearman, partial rank correlation); decision-analytic cost-effectiveness models

Reproducible research: R Markdown, Git and GitHub, structured data management

Laboratory techniques: indirect ELISA, DNA extraction, PCR, microscopy, histopathology

Geospatial analysis: QGIS (spatial data visualization and disease mapping)

Scientific visualization: ggplot2, Adobe Illustrator, Adobe Photoshop, BioRender, Canva

Productivity & languages: Microsoft Office; English (medium of instruction throughout university)

AWARDS, FELLOWSHIPS & HONORS

- **National Science and Technology (NST) Fellowship** — Ministry of Science and Technology, Bangladesh (2026), awarded for MS thesis research.
- **Aspire Leaders Program — Finalist (2025).** Selected from more than 36,000 applicants; completed 40 hours of leadership coursework from the Aspire Institute (founded by Harvard Business School faculty).
- **Education Board Merit Scholarship** — Secondary School Certificate (2015) and Higher Secondary Certificate (2017).

CERTIFICATIONS & PROFESSIONAL TRAINING

- **Veterinary Internship** — BAU (Jul – Dec 2023): clinical, farm and wildlife settings.
- R Programming for Research — Udemy (2024).
- Agricultural Extension Education field training (6 days) — BAU (2024).

PROFESSIONAL MEMBERSHIPS & SERVICE

- **Peer reviewer —** BMC Infectious Diseases; Evaluation and Program Planning
- **Public engagement —** Findings on brucellosis risk factors and milk-production losses in Bangladeshi dairy herds covered by Somoy TV (national television) and national news outlets, 2026
- Member, Japan Epidemiological Association — Sep 2026 – Present
- Student Member, American Society for Microbiology (Global Outreach) — Jun 2024 – Present
- Member, Badhon (Voluntary Blood Donors' Organization) — Feb 2019 – Present

REFERENCES

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